Dams mark drought vulnerability rather than buffer it, and storage cannot offset a falling water supply in Chile

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Abstract

Expanding reservoir storage is the reflexive response to intensifying drought, yet whether dams build resilience or merely mark where vulnerability already concentrates has remained untested for want of a credible counterfactual. We treat reservoir presence as a basin-level intervention in Chile's ongoing megadrought, comparing how matched dammed and undammed basins respond to the same meteorological deficit. Across streamflow drought transmission, cropland change, and consumptive water rights, the reservoir effect brackets zero once siting is matched out. The cleanest test compares regulated reaches below each dam with unregulated reaches above: the apparent buffering is as strong upstream as downstream, the signature of aridity, not regulation, and a positive control confirms the design would have detected a real effect. Meanwhile the storage band drifts downward, indicating a falling supply. At the multi-year timescale reservoirs mark vulnerability established before our record and added none through the megadrought; adaptation must shift toward managing demand.

Keywords: reservoir effect, drought propagation, socio-hydrology, causal inference, megadrought, Chile

As the world dries under warming that is intensifying drought across many regions[1,2], governments instinctively build more dams. Reservoirs are the dominant structural response to drought, and in the short term they work, more storage raises water availability and lowers the frequency, severity and duration of shortage[3]. Yet a socio-hydrological view warns that the same infrastructure can be self-defeating over a dam's planning horizon. Greater supply invites agricultural, urban and industrial expansion, so demand grows to meet and then exceed each new increment of storage (the supply-demand cycle); and prolonged reliance on abundant stored water erodes the incentive to adapt, deepening dependence and magnifying the damage when a drought finally outlasts the reservoir (the reservoir effect)[3]. Consistent with this, global

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water demand has outpaced storage for decades, the storage returns of new dams are diminishing worldwide[4], and most reservoirs now fail to refill under intensifying drought[5]. Whether these feedbacks dominate, or are outweighed by the protection dams provide, remains debated[6].

The obstacle is that the reservoir effect for drought has been described far more than it has been measured. Reservoirs are not sited at random, they are built where irrigation demand and aridity already concentrate, so dammed basins differ systematically from undammed ones before any drought strikes. The short-term hydrological benefit is documented, as regulation shortens downstream hydrological drought and lengthens the lag from meteorological drought[7,8], though it can also redistribute scarcity in space and time[9]. But the long-term, demand side is far less constrained, and separating a genuine reservoir effect from the siting that accompanies it requires a defensible counterfactual, namely what a drought's impact *would have been* in a basin had the reservoir not been built[3]. Prior dammed-versus-undammed and before-after comparisons cannot supply it.

Chile is an acute testbed. Since 2010 its central regions have endured a *megadrought*, an uninterrupted run of dry years now shading into structural aridification as precipitation falls and atmospheric evaporative demand rises, a drying with a documented anthropogenic component[10–14]. Because the 1981 Water Code makes water-use rights perpetual private property that the state cannot readily curtail, the policy reflex has been supply-side: new dams and inter-basin transfers, alongside market incentives that concentrated water in high-value export crops[15,16]. Chile thus exhibits, in compressed form, every ingredient of the reservoir effect: an extreme and persistent forcing, a supply-side response, and demand expansion under weak adaptive feedback.

Here we ask whether reservoirs build resilience or merely mark, and postpone, vulnerability under prolonged drought. We treat the presence of a major reservoir as a basin-level intervention and construct the missing counterfactual in two steps. First, we compare each dammed basin only against undammed basins that share its climate and landscape, and condition on the drought actually experienced, asking whether dammed and comparable undammed basins convert the same meteorological deficit into different impacts. Second, within each dammed basin we compare regulated reaches below the dam against physically unregulated reaches above it, which share climate, geology and siting but not the reservoir. Together these turn the socio-hydrological reservoir-effect hypothesis into a falsifiable, causally framed test, in the setting where it matters most.

Results

Reservoirs sit where water is already scarce

Reservoirs in Chile are built where drought bites hardest: our 21 dammed basins cluster in the arid centre and north, so a raw comparison of dammed against undammed basins would confuse the reservoir with its location. We separate the two with the design in Figure 1. First, we compare each dammed basin only with undammed basins that share its climate, size and elevation, statistical matching that keeps 21 of 24 dammed basins against an effective 91 of 244 controls (Supplementary Figs. S4 and S5), and we condition on the drought each basin actually experienced, comparing impact per unit of meteorological deficit rather than raw levels. Second, within each dammed basin we compare regulated reaches below the dam against unregulated reaches above it. A genuine reservoir effect must show up in these comparisons; in none of them does it.

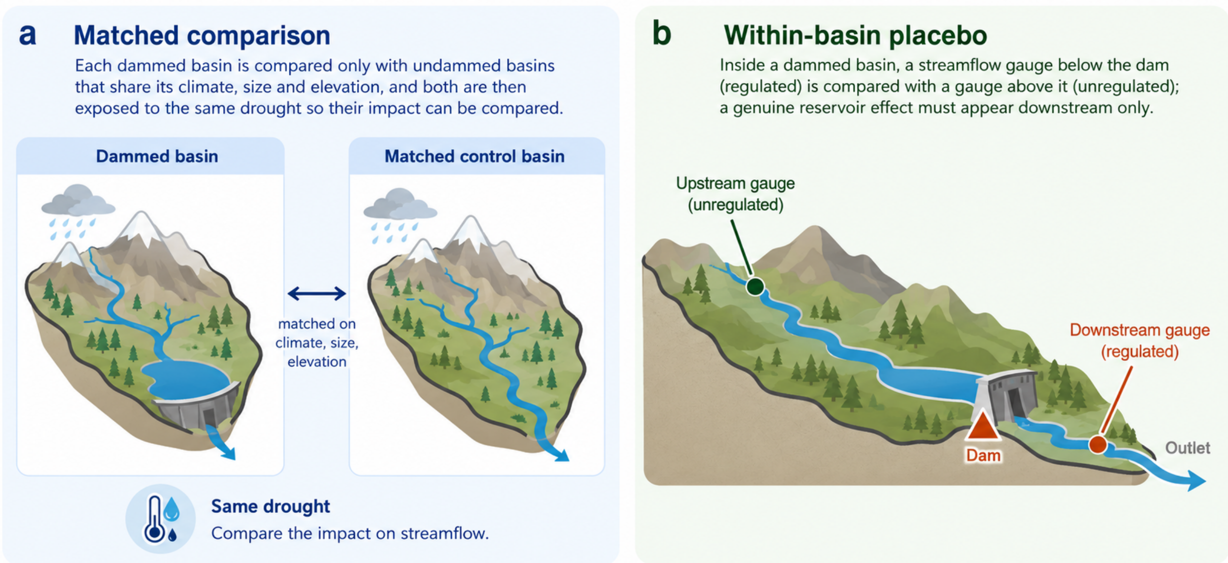

Figure 1: The two comparisons that separate the reservoir from its location. (a) Matched comparison: each dammed basin is compared only with undammed basins that share its climate, size and elevation, and both are then exposed to the same drought so their impact can be compared. (b) Within-basin placebo: inside a dammed basin, a streamflow gauge below the dam (regulated) is compared with a gauge above it (unregulated); a genuine reservoir effect must appear downstream only. Schematic; the full study-area map and matching diagnostics are in Supplementary Figs. S4 and S5.

The apparent buffering is the location, not the dam

The most direct and intuitive test is on streamflow. Meteorological drought (Standardized Precipitation Evapotranspiration Index of 12 months, SPEI-12) propagates into streamflow drought (Standardized Streamflow Index of 12 months, SSI-12) more gently in dammed basins, the visual signature of buffering (Figure 2, panel a). But this flattening tracks where dams sit, not what they do. If the dam were doing the buffering, transmission would flatten only below it; instead it flattens just as much at the unregulated *upstream* reaches of the same basins (differential slope −0.20 upstream versus −0.16 downstream; Figure 2, panel b). Because the upstream line flattens too, the cause must be the aridity of the location, not the reservoir. The national slope gap of −0.18 accordingly collapses to about −0.05 once dammed and control basins are compared within the same climate region, and in the high-supply south, where buffering should be easiest to see, the upstream and downstream gaps are statistically identical (−0.036 versus −0.034). None survives randomization inference ($p_{\text{perm}} = 0.39$). The apparent buffering is therefore the between-region aridity gradient, the textbook signature of a siting confound, not an effect of regulation; a step-by-step decomposition confirms the signal survives matching and fixed effects and dies only when each climate region is allowed its own baseline transmission (Supplementary Fig. S3 and Table S4). Upstream gauges do sit higher, and elevation predicts transmission among controls, but elevation cannot manufacture a downstream-specific effect. The 0.54 km up/down elevation gap times the control sensitivity (−0.21 per km) already over-explains the observed −0.04 difference, leaving a residual downstream-specific buffering of at most about 0.07 (13% of the 0.53 downstream-panel baseline; the intent-to-treat baseline used for the equivalence margin is 0.585), well below the 46 to 59% of baseline the design can detect (Supplementary Table S2).

No reservoir effect survives across any operational pathway

The same answer holds across the other pathways by which reservoirs are thought to modify drought (Figure 3, Table 1). We measure each effect as the differential response of dammed and matched-control basins to the same meteorological deficit, and place them on a common axis. Each estimate is divided by its own standard error, so effects on different scales can be read on one plot, and a value inside ±2 is indistinguishable from no effect. Every interpretable estimate brackets zero: cross-sectional cropland-area expansion (−0.69 ha km^{-2}, 95% CI $[-2.4, 1.1]$) and evapotranspiration buffering (−0.014 $[-0.065, 0.037]$), and the forcing-conditioned slope gaps for cropland area ($p_{\text{perm}} = 0.91$) and orchard evapotranspiration ($p_{\text{perm}} = 0.24$). One outcome, whole-basin evapotranspiration, appears positive, but its pre-trends fail and it is confounded by residual aridity over barren land, so we flag it as uninterpretable rather than fold it into the nulls; at face value it points toward *vulnerability*, not buffering. Restricting evapotranspiration to vegetated (irrigated orchard) cells was a post-hoc diagnostic prompted by that failure: masking barren pixels removes both the pre-trend failure (p from 3×10^{-5} to 0.40) and the positive effect (+0.048 to −0.020), consistent with barren-land aridity contamination rather than a reservoir effect (Supplementary Fig. S7); because the stratum was defined after inspecting the whole-basin result, we rest no claim on it.

This null is informative, not merely underpowered, but the power is concentrated in the streamflow pathway.

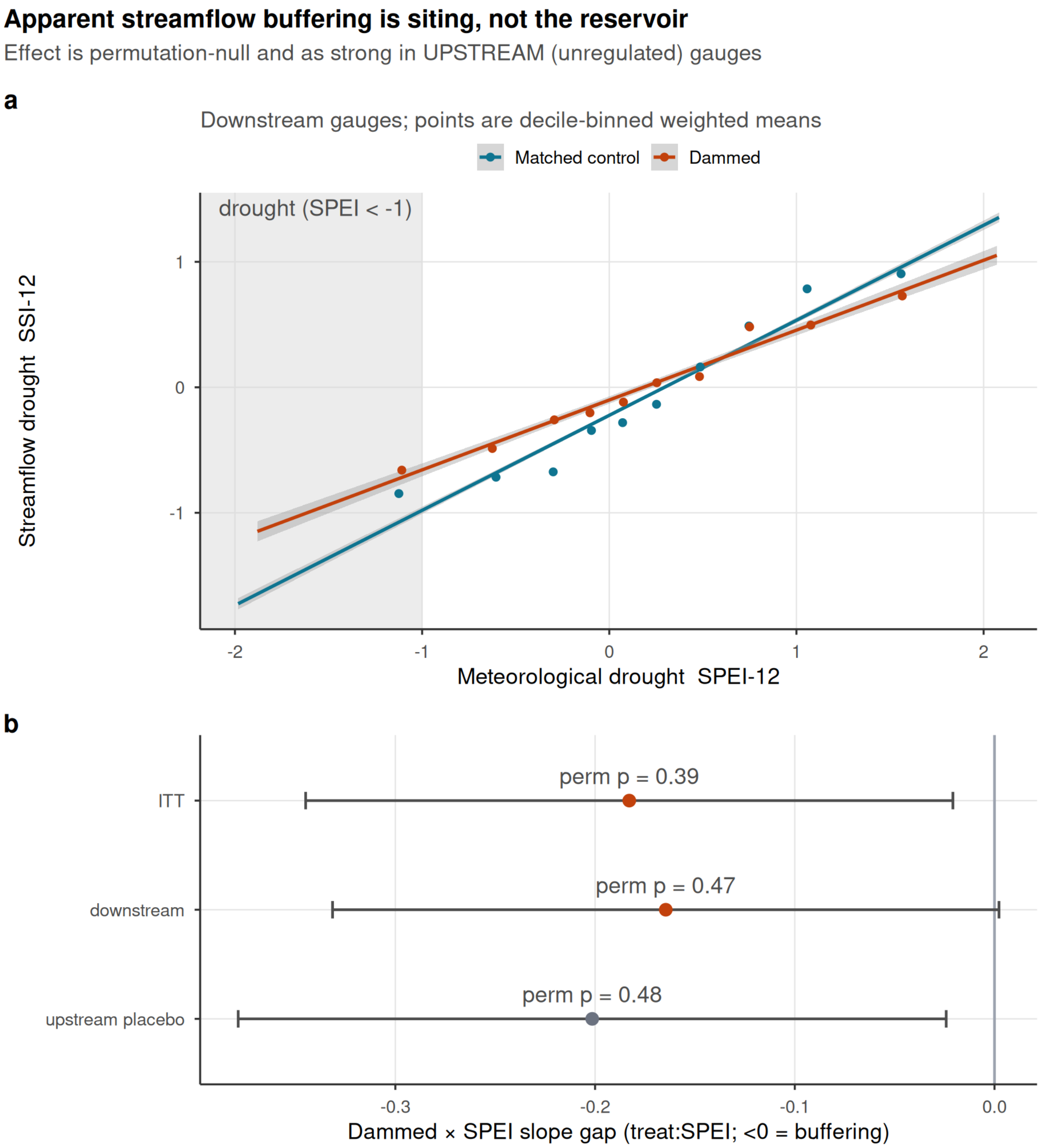

Figure 2: Streamflow buffering is reservoir siting, not regulation. (a) SPEI-12 to SSI-12 transmission for dammed versus matched-control basins (downstream gauges); lines are weighted least-squares fits and shaded bands are 95% confidence intervals; dammed basins show a flatter slope, the apparent buffering. (b) Differential slope (treat:SPEI; <0 = buffering) for intent-to-treat, downstream-only, and the upstream-only placebo; horizontal bars are 95% confidence intervals (estimate +/- 1.96 cluster-robust SE), annotated with randomization-inference p-values. All are permutation-null, and the upstream placebo (above-dam, unregulated) is as strong as downstream, so the attenuation is siting/aridity, not the dam.

With about 21 dammed clusters the design cannot claim exact equivalence to zero, but it rules out streamflow buffering larger than 46 to 59% of baseline transmission, and a positive control confirms the point: injecting a known buffering effect into the treated gauges, the estimator recovers it one-to-one and randomization inference detects it beyond that threshold (Methods; Supplementary Fig. S2 and Tables S1, S3). The demand-side designs are individually far weaker: the minimum detectable effect is about 237% of baseline for orchard-evapotranspiration transmission and orders of magnitude above baseline for the cropland-area slope gap (Supplementary Table S2), so those nulls corroborate the streamflow result rather than independently bounding their own pathways.

Table 1: Reservoir-effect estimates across estimators. Cross-sectional doubly-robust matched ATTs (including the water-rights induced-demand test) and forcing-interacted DiD slope gaps (treat:SPEI). The DiD rows and water-rights accrual are judged by randomization-inference p-values (p), the design's primary inference at ~21 clusters; cluster-robust CIs are shown for completeness and over-reject here. A positive estimate would indicate induced-demand vulnerability; a negative one, operational buffering. *Whole-basin ET fails its event-study pre-trend test (differential pre-2010 trend p approximately 3e-5): its cluster-robust CI excludes zero but the panel is confounded by barren-land aridity, so it is labelled confounded and excluded from the equivalence bounds (Supplementary Table S1).

Group	Outcome	Estimand	Estimate	95% CI	p	Verdict
Cross-sectional matched ATT	Cropland-area expansion	ha km^-2 added	-0.687	[-2.44, 1.06]	-	null
Cross-sectional matched ATT	Reservoir ET buffering	d log ET / d SPEI	-0.014	[-0.0646, 0.0369]	-	null
Cross-sectional matched ATT	Water-rights accrual	rights / 100 km^2 added	20.036	[4.02, 36]	0.52	null
Forcing-interacted DiD (slope-gap)	Cropland area	differential deficit->impact slope	0.001	[-0.00344, 0.00493]	0.91	null
Forcing-interacted DiD (slope-gap)	Whole-basin ET	differential deficit->impact slope	0.048	[0.0255, 0.0703]	0.11	confounded*
Forcing-interacted DiD (slope-gap)	Orchard ET	differential deficit->impact slope	-0.020	[-0.0448, 0.00565]	0.24	null

Reservoirs mark prior vulnerability; the megadrought added none

Dammed basins do carry the vulnerability reservoirs are blamed for concentrating: they hold about 4.6 times more cropland than matched controls (11.1% versus 2.4% of basin area; Figure 4, panel a). But this gap is a *siting* signature, present and stable across the 2000–2024 record, with dammed and control trajectories evolving in parallel through the megadrought. A dynamic event study confirms it: the differential trend is flat before 2010 (cluster-robust $p = 0.65$, permutation $p = 0.72$) and does not diverge afterwards ($p = 0.81$; Figure 4, panel b). The flat pre-trend meets the same power standard as the headline nulls: the design would detect a pre-2010 divergence accumulating to about half the control cropland level by 2009, and the observed trend is a seventh of that minimum detectable slope (Supplementary Table S14). Reservoirs mark where cropland already was; they neither expanded it nor changed how it responds to drought.

The same holds for the most direct measure of demand. In the Chilean Water Directorate (DGA) registry of consumptive water rights, dammed basins do appear to accrue rights faster than controls through the

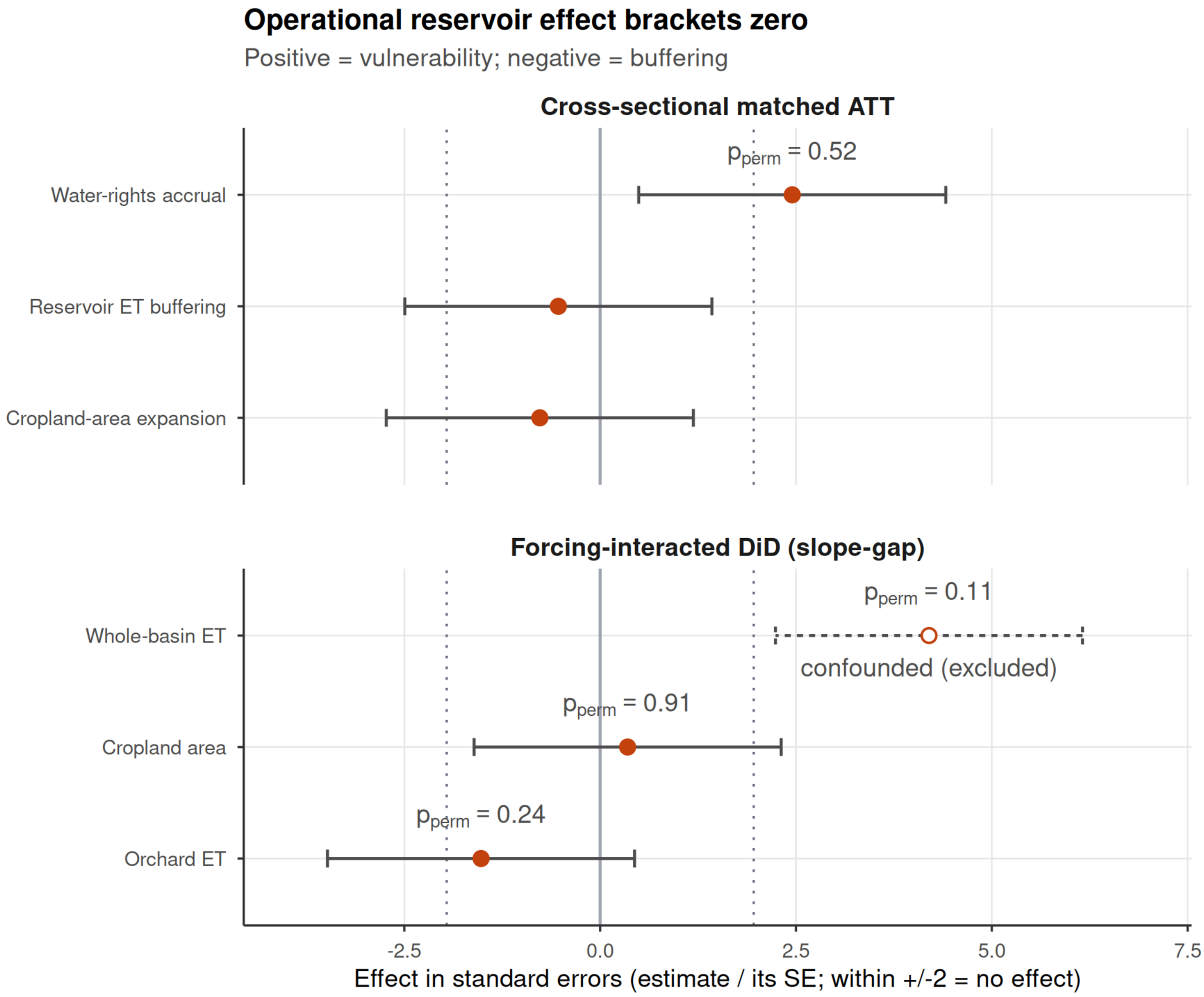

Figure 3: Dammed-versus-control effect across the estimators, each expressed uniformly in its own standard errors (estimate divided by its SE) so the axis metric is identical for every row; an estimate inside +/-2 is indistinguishable from no effect. Rows are grouped by inference regime: cross-sectional matched ATTs (top) and forcing-interacted DiD slope gaps (bottom). Bars span +/-1.96 SE (the 95% reference interval) and dotted lines mark +/-1.96; a positive effect would indicate induced-demand vulnerability, a negative one operational buffering. Rows judged by randomization inference are annotated with their p (the design's valid inference at ~21 clusters, since the +/-1.96 SE guides over-reject there). Under that valid inference every estimate brackets zero. Two rows sit outside +/-1.96 by their naive standard error yet are null by the design: water-rights accrual (p_perm = 0.52; the apparent induced demand lies inside the permutation null expected from the dammed basins' unbalanced aridity, Supplementary Fig. S6) and whole-basin ET (open symbol, dashed interval), which additionally fails its event-study pre-trend test and is confounded by barren-land aridity, so it is labelled confounded and excluded (Table 1).

megadrought, on both the outlier-robust count (+20 rights per 100 km²; Figure 3, Table 1) and the winsorized volume (+2.5 l/s per km²; Supplementary Fig. S6), each with an analytic interval excluding zero. This is the naive-analysis trap the design exists to catch: dammed basins are more arid, arid basins carry denser water-rights activity, and this imbalance drives the gap. The count does not survive randomization inference ($p_{\mathrm{perm}} = 0.52$), lying inside the null that random assignment within climate and aridity strata produces (Supplementary Fig. S6), and remains null under a fully non-parametric match on the aridity-overlap subset ($p_{\mathrm{perm}} = 0.42$; Supplementary Table S7). The winsorized volume is more delicate: stable across winsorization thresholds (Supplementary Table S8), its overlap-subset permutation p falls to 0.005, but that signal is an artefact of spatial dependence (residual Moran's $I = 0.39$; Supplementary Table S9); under a spatially-restricted permutation that swaps treatment at the cuenca-block level it is no longer significant ($p_{\mathrm{perm}} = 0.077$) and its cuenca-clustered interval spans zero ($[-0.3, 4.1]$). Even on water rights, then, no induced demand is robustly attributable to the reservoir once siting and spatial structure are accounted for; a conventional or spatially-naive matched analysis would have concluded otherwise.

Storage falls; a change in refill amplitude cannot be resolved

If reservoirs neither buffer nor amplify drought, the change that matters under the megadrought is in how much water there is to store (Figure 5). Over 2005–2024 the entire storage band drifts downward in step: annual peak storage falls by 1.3 percentage points of capacity per year ($p = 0.013$) and the trough almost identically (1.2 pp yr^{-1}, $p < 10^{-4}$), while the seasonal amplitude shows no detectable trend (-0.06 pp yr^{-1}, $p = 0.91$; Supplementary Table S5). That amplitude interval is wide: the data cannot determine whether refill amplitude, the buffering capacity, has degraded, and we draw no conclusion about it; what the band shows unambiguously is the decline of both peak and trough. That decline is commensurate with the climate-driven supply reduction measured independently in the matched controls: unregulated control streamflow fell about 2.9% per year over 2005–2020 (dammed basins 3.4%), a primary estimate from our gauge panel consistent with documented megadrought runoff deficits[10,17], and comparable to the ~1.4% per year decline in reservoir peak storage. The comparison is correlative rather than a water-balance attribution (Discussion), but the storage decline is at least consistent with, and no larger than, the falling supply seen in unregulated flow. Our direct demand outcomes point the same way: cropland area in dammed basins did not expand (Figure 4, Table 1), and neither did consumptive water-rights allocation (Supplementary Fig. S6)[4,5].

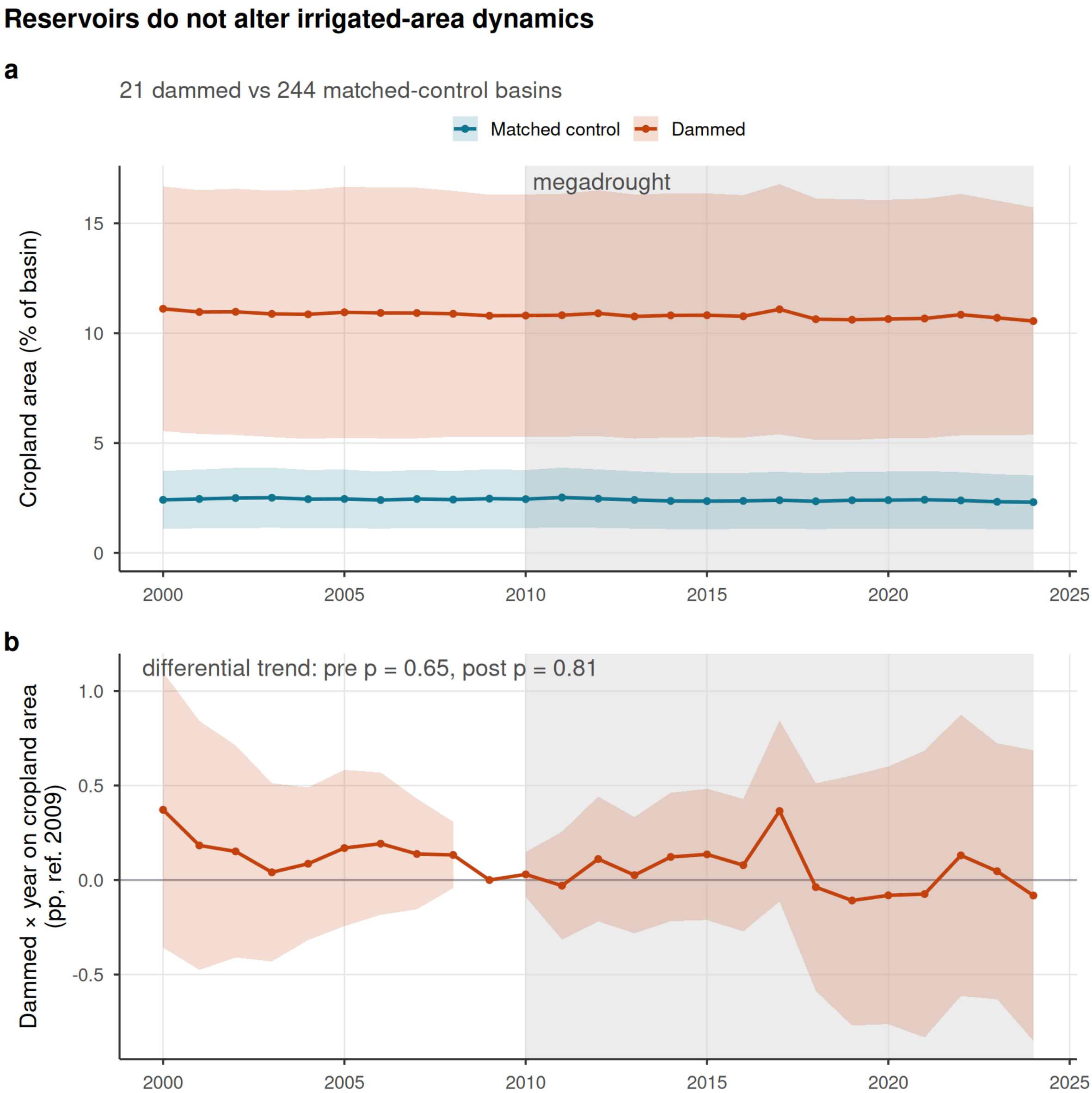

Figure 4: Observed cropland-area dynamics in dammed versus matched-control basins (21 dammed, 244 controls). (a) Entropy-balancing-weighted mean cropland area (% of basin, MapBiomas class 18) with 95% bands (weighted mean +/- 1.96 SE across basins); the y-axis is anchored at zero. Dammed basins hold ~4.6x more cropland (11.1% vs 2.4% of basin area) but the trajectories evolve in parallel. (b) Dynamic event study (dammed x year, reference 2009): the shaded band is the 95% confidence interval (coefficient +/- 1.96 cluster-robust SE). The differential pre-trend is flat (pre-2010 trend-slope cluster-robust p = 0.65, permutation p = 0.72) and does not diverge afterwards (post p = 0.81); a linear trend-slope test is used because the joint event-study Wald over-rejects at ~21 treated clusters. The 2010-2024 megadrought is shaded in both panels.

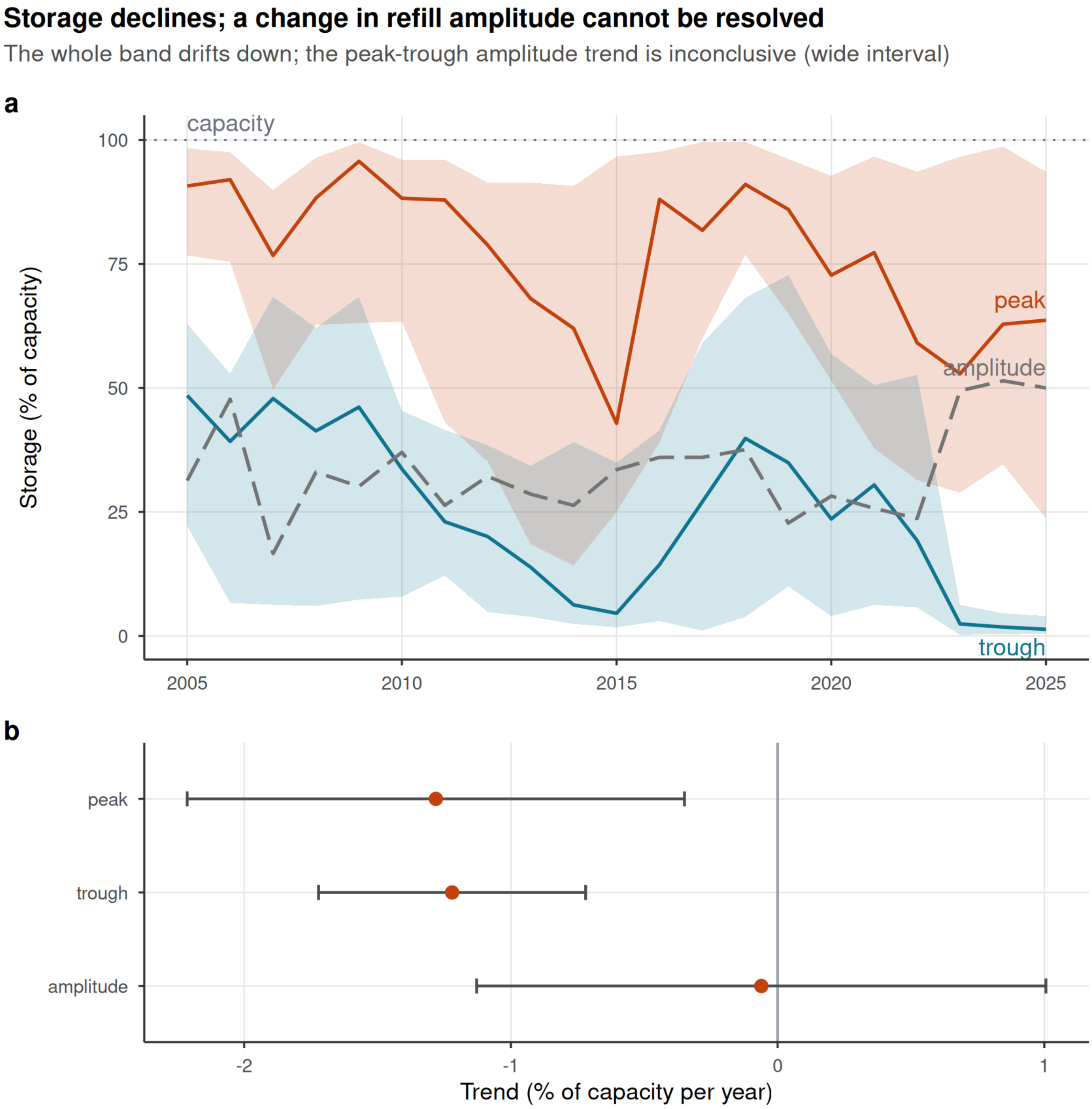

Figure 5: Reservoir storage declines; a change in seasonal refill amplitude cannot be resolved. (a) Cross-reservoir annual storage band: realized median peak and trough percent-of-capacity (solid lines) with interquartile ribbons across reservoirs, and the peak-trough amplitude (grey dashed). Medians are used because a few reservoirs report storage above nominal capacity (dotted 100% line); the whole band drifts down over 2005-2024 while the amplitude does not narrow. Central lines are the realized medians, not linear fits; the net trend is quantified in (b). (b) Per-year trend slope of peak, trough, and amplitude (reservoir fixed effects); error bars are 95% confidence intervals (estimate +/- 1.96 cluster-robust SE): peak and trough both decline while the amplitude trend is not distinguishable from zero (a wide interval, so statistically unchanged rather than proven constant), consistent with a supply-side level decline rather than refill/buffer degradation. Descriptive: pooled within-reservoir trend, no control group.

Discussion

Across a matched national design and four independent estimators, the operational effect of reservoirs on drought brackets zero once siting is matched out. Dammed basins do hold about 4.6 times more cropland than matched controls, but that gap is fixed across the record, with parallel pre-trends and no megadrought divergence. The evidence supports one reading: reservoirs mark where water-dependent vulnerability already concentrated, in arid, already-cultivated basins, rather than make it. This is a finding, not a failure of power: a credible cause of the apparent effect is identified, and the positive control shows the design would detect genuine buffering. The within-basin placebo is the cleanest falsification: a regulation effect must appear below the dam and be absent above it, and it is not; what remains after the placebo and within-region conditioning is the aridity gradient itself.

The corollary is transferable: naive dammed-versus-undammed or before-after contrasts inherit the full siting confound[7,9,18,19], and the upstream placebo applies to any regulated river with gauges above and below the structure.

If reservoirs neither buffer nor amplify multi-year drought operationally, what matters under a multi-decadal dry spell is the water available to store. The storage band drifts downward, peak and trough in step at 1.3 and 1.2 percentage points of capacity per year, while the amplitude trend is too imprecise to determine whether buffering capacity has degraded. The powered part of the record is the level decline, which points to a falling supply ceiling. The trend is descriptive, confounded with secular changes in management or unrecorded abstraction, so we make no causal attribution, but it is commensurate with the independent decline in unregulated control streamflow. Read against Chile's documented transition from episodic drought toward structural aridification[10,12,17] and the global pattern of reservoirs failing to refill[4,5], the binding constraint is shifting toward water supply rather than storage capacity.

These findings give the socio-hydrological *reservoir effect* and *supply-demand cycle*, contested for want of quantification[3,6], a credibly identified test, and find the protective function over-attributed to storage. Because treatment is time-invariant during a continuous drought, we test the spatial imprint of these feedbacks, not the longitudinal coevolutionary loop: the 4.6-fold cropland gap could be the equilibrated legacy of demand induced at construction, before our record, so our null scopes to the megadrought era rather than ruling out historical induced demand, which would need staggered commissioning to test. Within that scope the nulls refine the theory rather than refute it: the supply-demand cycle leaves no operational imprint during drought, suggesting its mechanisms act through siting and construction rather than through operation, a distinction current models do not draw. The findings parallel the *efficiency paradox* in irrigation[20]; the common lesson, that supply-side infrastructure does not resolve scarcity driven by demand or falling supply[21], is strongest on the supply side. How far the demand-side null travels depends on governance: the siting lesson should hold wherever dams are placed non-randomly, but prior-appropriation systems, where unused rights risk forfeiture, may amplify induced demand, while state-managed adaptive allocation can curtail it, so the reservoir effect is governance-mediated, not universal.

The policy implication is direct. Chile's 1981 Water Code makes water-use rights perpetual private property, so the reflexive response has been supply-side, new dams and inter-basin transfers[11,15]; the 2022 reform moves governance toward the demand-side, adaptive allocation our results argue for[22], though it postdates our window. Under structural aridification, building more storage is a supply-blind response to a supply problem: the buffering benefit dams provide is a property of where they sit rather than of what they do[23]. Adaptation must shift toward demand management, reallocation and managed aquifer recharge[20,24].

Several limitations bound these conclusions, each tested and quantified in the Supplementary Information. The result is a bounded null, not a proven equivalence, the design excludes streamflow buffering larger than 46 to 59% of baseline transmission, and the demand-side outcomes are far less powered (minimum detectable effects near 237% of baseline for orchard-evapotranspiration transmission and far above baseline for the cropland slope gap; Supplementary Table S2), so the causal weight rests on the placebo and the demand-side nulls are corroborating rather than independently decisive. Groundwater substitution is bounded, despite the region's unsustainable aquifer drawdown[25], dammed and control water tables fall at indistinguishable rates across 213 DGA wells, though monitored wells need not sample the orchard footprints or strata where pumping concentrates (Supplementary Table S10). The snowmelt confound in the placebo, acute in these snow-fed Andean basins[26,27], is tested on the differential estimand itself: the up/down equivalence survives a forcing-by-snowpack adjustment, persists among low-snow units, does not strengthen in high-snow years, and is statistically indistinguishable between early and late halves of the panel (period interaction $p = 0.46$ upstream, 0.52 downstream; Supplementary Table S12). Administrative closure cannot force the demand null, treated basins entered the drought with higher allocated stock yet accrued new rights 2.7 times faster than controls, and the parallel control cropland trajectory contradicts the collapse-prevention counterfactual (Supplementary Table S15). No matched pair shares a watershed; transfers out of dammed basins would shrink the estimated gap, and imports into them would mimic downstream-specific buffering, which is absent (Supplementary Table S15); restricting to irrigation-only reservoirs leaves the result unchanged (Supplementary Table S11). MOD16 (the Moderate Resolution Imaging Spectroradiometer evapotranspiration product) underestimates evapotranspiration (ET) over irrigated land, biasing the orchard-ET gap toward zero; that null may be an artifact of the product and carries no weight on its own; the demand conclusion rests on cropland and the water-rights registry, which do not share the bias, and a thermal ET sensitivity (Landsat or ECOSTRESS class) remains future work. Finally, the 12-month accumulation targets multi-year vulnerability, not the sub-seasonal buffering reservoirs are also built to provide, which we neither test nor dispute. A rule-curve quasi-experiment on release records, once public, is the priority for future work.

Returning to the motivating question, at the multi-year timescale of structural aridification reservoirs neither create resilience nor postpone vulnerability operationally: storage neither buffers nor amplifies the drought signal beyond what siting implies, while the water available to store declines. Resilience to a falling supply ceiling will come from confronting demand and allocation, not from more impoundments.

Methods

Study area and design

We study continental Chile (18–44° S), a strong north–south aridity gradient spanning hyper-arid, Mediterranean, and humid-temperate climates, over 2005–2024, the window in which reservoir storage, meteorological forcing, and satellite outcomes jointly exist. The design is a matched comparison of dammed versus undammed sub-watersheds, with reservoir presence treated as a basin-level intervention. Because the reservoirs in most treated basins (18 of the 24) were commissioned before the storage record begins, treatment is effectively time-invariant over the window and a staggered-adoption (commissioning) event study is infeasible; identification therefore rests on a matched cross-section combined with conditioning on meteorological forcing rather than on calendar time.

The central estimand is the operational effect of a reservoir on drought outcomes *conditional on the covariates that govern where reservoirs are sited*, that is, whether dammed basins respond to the same drought forcing differently from hydroclimatically comparable undammed basins. This isolates reservoir *operation* from reservoir *siting*[3].

Spatial units and treatment assignment

The analysis grain is the DGA sub-watershed (467 units), chosen over the coarser watershed grain (101 units) for power and common support, reservoirs sit in Chile's largest main-stem basins, and the finer grain supplies far more controls inside the treated size range (detailed in the archived analysis repository). A sub-watershed is treated if it contains 1 of the 26 monitored reservoirs by point-in-polygon assignment, giving 24 treated units. The reservoirs are predominantly irrigation (20 of 26; three hydropower, two potable supply, one mixed), so the treated set is dominated by the use type built to buffer agricultural drought, and the headline streamflow result is unchanged when restricted to irrigation-only treated basins (Supplementary Table S11). An untreated sub-watershed that lies within a watershed containing a reservoir is flagged as contaminated (it shares the regulated main stem and is not a clean counterfactual) and removed (45 units), leaving 398 clean controls. This contamination rule defuses the most direct downstream-spillover (SUTVA: Stable Unit Treatment Value Assumption) violation; residual adjacency and shared-aquifer spillovers between neighbouring sub-watershed are carried as an identification caveat and checked at watershed grain.

Data

Reservoir storage and treatment. Monthly storage for 26 monitored reservoirs (the DGA record runs January 2005 to April 2026; we analyse the complete calendar years 2005–2024) with a per-reservoir capacity column (maximum level in hm^3); storage and capacity share units (hm^3), so percent-of-capacity (`100 × level / maximum level`) is the cross-reservoir-comparable state variable. Static attributes (commissioning year, use, size class) come from the national dam registry.

Meteorological forcing. The Standardized Precipitation-Evapotranspiration Index (SPEI) at a 12-month accumulation, derived from Climate Hazards Center InfraRed Precipitation version 3 (CHIRPS) and Climate Hazards Center InfraRed Temperature with Stations (CHIRTS) and standardized on the 1991–2020 climatological baseline[28], is the primary exogenous dose; CHIRPS-derived standardized indices are validated for Chilean drought monitoring, including ENSO-modulated variability across the country[29–31]. We use the 12-month accumulation because it matches the annual, snowmelt-fed hydrological cycle of central Chile (winter precipitation stored as snow and released through the growing season) and so captures the inter-seasonal carryover that irrigation reservoirs are built to bridge, at the scale at which multi-year vulnerability (rather than sub-seasonal operation) is expressed. The treatment estimates are insensitive to this choice: re-estimating the forcing-conditioned effect with 6- and 24-month accumulations leaves it materially unchanged (archived analysis code). By construction the 12-month window does not resolve short-term (1–3 month) operational buffering, which reservoirs are documented to provide[7,8] and which we flag as a boundary of our conclusions (Discussion). Conditioning on forcing (estimating effects on the deficit-to-impact *slope* rather than on levels) is how we break the collinearity between the 2010-onset megadrought and the treatment clock.

Hydrological drought. Daily streamflow from the CR2 network (2000–2020) is aggregated to the Standardized Streamflow Index at 12 months [SSI-12;[32];[33]], the regulated-flow analogue of SPEI-12 and the index in which meteorological deficits are re-expressed as hydrological drought[34], used as the outcome in the within-basin placebo (below). Gauges are classified as upstream or downstream of each dam from an SRTM digital elevation model.

Ecological and land-use outcomes. (i) annual cropland area from MapBiomas Chile (https://chile.mapbiomas.org/) Collection 2 (30 m, 1999–2024); (ii) actual evapotranspiration from MODIS MOD16 (8-day, ~500 m, 2001–2024), for whole basins and for an orchard stratum delineated from the CIREN/ODEPA cadastre (https://icet.odepa.gob.cl/). Satellite vegetation and evapotranspiration signals track drought-driven losses of agricultural productivity in Chile closely enough to serve as impact outcomes[35].

Water rights. The DGA consolidated registry of water-use rights (*Derechos de Aprovechamiento de Aguas*), giving for each right its grant date, consumptive/non-consumptive type, surface/groundwater nature, use, allocated flow (harmonized to l/s), and UTM capture point. We assign consumptive rights to sub-watershed by point-in-polygon and build a per-basin annual panel of the cumulative allocated-rights stock, our direct demand-side outcome. Because the registry contains severe volume errors (a single physically impossible entry can dominate a basin's total, e.g. a caudal exceeding 200,000 m^3/s), the outlier-robust primary measure is rights *density* (count per 100 km^2); a volume series in which each individual consumptive flow is winsorized at the 99.5th percentile of the consumptive-flow distribution gives the same result, and the volume null is insensitive to the winsorization threshold across the 95th, 99th, 99.5th and 99.9th percentiles (permutation p from 0.39 to 0.21; Supplementary Table S8). Both are reported (Supplementary Fig. S6).

Snow water equivalent. ERA5-Land monthly snow-water-equivalent (SWE, 0.1°, 2000–2026), reduced to per-gauge and per-sub-watershed peak-SWE climatologies plus annual peak-SWE anomalies, to test the snowmelt confound on the placebo estimand itself (Supplementary Table S12): we refit the up- and downstream buffering models with a forcing-by-snowpack adjustment, within the low-snow half of units (below-median peak SWE), across early and late halves of the panel, and with a snow-year interaction that asks whether the apparent buffering strengthens in high-snow years. ERA5-Land underestimates absolute Andean snowpack, the same class of product caveat we apply to MOD16 evapotranspiration, so all snow terms enter standardized, which is insensitive to multiplicative bias; gauge-level sensitivity checks among undammed controls are likewise reported per standard deviation of peak SWE.

Groundwater levels. DGA well hydrographs of depth to water (2000–2021; 216 wells carry usable hydrographs nationally, of which 213 fall inside the matched basins and enter the bound; a Hampel filter removes measurement spikes), assigned to sub-watershed by point-in-polygon from well coordinates. Each well's megadrought (2010–2021) depletion rate is the OLS slope of depth to water on time (m yr^{-1}, positive = falling water table); per-basin values aggregate well rates by their outlier-robust median. These bound the groundwater-substitution confound as a matched dammed-versus-control Average Treatment Effect on the Treated (ATT) on the depletion rate (Supplementary Table S10).

Matching covariates. Baseline aridity (long-term mean annual precipitation divided by atmospheric evaporative demand P/AED over the 1991–2020 World Meteorological Organization normal), watershed area, mean elevation (SRTM), and Köppen–Geiger climate class, extracted per sub-watershed by zonal statistics. Because the 1991–2020 normal overlaps eleven megadrought years, baseline aridity could in principle absorb post-treatment variation; we verify the window is innocuous. Unit aridity recomputed on the strictly pre-drought 1991–2009 window is essentially identical to the full-window covariate (Pearson $r = 0.998$, Spearman $\rho = 0.999$ across the 265 matched units, and 99% of units keep the same aridity tercile used for the permutation strata), and the megadrought change in P/PET does not differ significantly between treated and weighted control basins (-0.04, $p = 0.10$, an order of magnitude smaller than the ~0.5 treated-control aridity gap the matching addresses), so the matching does not lean on drought-period or reservoir-influenced variation (Supplementary Table S13).

Matched control construction

We estimate the average treatment effect on the treated (ATT) using **entropy balancing** (`WeightIt`, `method = "ebal"`) exact-matched within Köppen main group, balancing **log(area)** and **mean elevation** (see the archived analysis repository). Balancing on size and elevation rather than latitude is deliberate: latitude and the climate class fight each other across the Andes–Patagonia span, and substituting SRTM elevation raised the effective control sample size from 9 to 91 while balancing latitude for free.

The procedure retains 21 of 24 treated sub-watershed (three high-Andes units fall outside local elevation common support) and achieves essentially perfect balance (standardized mean differences for log(area) and elevation fall from 1.18 / 0.37 to 0.000, variance ratios 1), with an effective control sample size 91 out of 244 in-window controls. The 244 in-window controls are the subset of the 398 clean controls that survives the common-support trim shared by every matching estimator: a control must lie in a Köppen main group containing a treated unit and within 250 m of that group's treated elevation range, and groups with fewer than ten such controls are dropped, so 154 clean controls fall outside this support and never enter the matching pool. Baseline aridity is a genuine confound (dammed basins are markedly more arid: median P/PET 0.26 vs 0.80 unweighted) but is *not* used as a balance target, because hard-balancing it collapses the ESS from 91 to 19; instead the weighting reduces the aridity imbalance to a residual SMD 0.17 (still above the 0.1 target), which the doubly-robust outcome model absorbs. Because this residual is real, the outcome-model adjustment for aridity is *load-bearing* rather than cosmetic and extrapolates across a genuine regime gap: the reported cross-sectional ATTs use a **linear** log-aridity adjustment, adding a quadratic aridity term leaves the qualitative conclusions unchanged (archived robustness), and restricting to the aridity-overlap subset (all 21 treated basins and 158 of 244 controls fall inside it) leaves the streamflow transmission-slope gap essentially unchanged (-0.17 vs -0.18) while shrinking the water-rights ATT to a value that now straddles zero, so the parametric adjustment is not masking a true effect (Supplementary Table S6). To rule out that the nulls are artefacts of this parametric functional form, we additionally re-fit entropy balancing *within* the aridity-overlap subset and estimate every cross-sectional outcome with the weighting-only estimator, which carries no outcome-model aridity term and so performs no extrapolation across the regime gap. On this fully non-parametric matching specification, cropland-area expansion, orchard-ET buffering and the water-rights count all remain null (Supplementary Table S7); the winsorized water-rights volume, which behaves differently here, is treated under spatial inference below. Crucially, the study's decisive evidence, the within-basin upstream/downstream placebo, the within-region transmission-slope comparison, and randomization inference, does **not** use the outcome-model aridity adjustment at all, so the central nulls do not hinge on extrapolating across the aridity gap. Robustness to the matching method is assessed with coarsened exact matching (CEM) and 1:2 nearest-neighbour (Mahalanobis) matching on the identical common-support sample; entropy balancing and NN agree on all 21 treated, while CEM additionally prunes the 4 most extreme large/high basins, so all treatment effects are reported with and without those units.

Causal estimands and estimators

Doubly-robust matched ATT. Each outcome is estimated three ways on the same sample, comprising weighting-only (ebal-weighted difference in means), regression-only (covariate-adjusted g-computation), and the headline **doubly-robust** estimator (ebal weights *and* an outcome model in log(area), elevation, and log(aridity)), so functional-form sensitivity is visible. Continuous covariates are centred at the treated mean so the treatment coefficient reads as the ATT; standard errors are HC3-robust (ebal weights treated as fixed, the standard mildly anti-conservative simplification).

Forcing-conditioned transmission slope. To remove megadrought *exposure* (concentrated in exactly the arid basins where reservoirs sit), the outcome for the DR machinery is the per-basin **transmission slope**, the regression of an annual outcome (e.g. log evapotranspiration) on annual SPEI-12 over 2005–2024, a standardized, cross-basin drought-transmission elasticity. The vulnerability hypothesis predicts a *steeper* slope (more impact per unit deficit) in dammed basins.

The two cross-sectional ATTs in Table 1 share the doubly-robust matched machinery but differ in outcome transform, and only the second is a transmission slope. (i) *Cropland-area expansion* is a level-change outcome, the change in MapBiomas cropland area per basin over the window (ha km^{-2} added, as Table 1 states); its protection against differential megadrought exposure comes from the forcing-interacted difference-in-differences below, not from the slope transform. (ii) *Reservoir ET buffering* applies the transmission-slope machinery: the per-basin slope of log evapotranspiration on SPEI computed over **irrigated orchard cells** (MOD16 ET on the CIREN/ODEPA cadastre footprint), which measures directly whether consumptive water use over irrigated land responds more or less to deepening drought in dammed basins. A negative slope gap indicates buffering, a positive one induced-demand vulnerability; the outcome is null, so consumptive use per unit of irrigated land shows no differential change with the reservoir.

Forcing-interacted difference-in-differences. On the observed area and ET panels we fit

$$y_{it} = \beta\,(\mathrm{dammed}_i \times \mathrm{SPEI}_{it}) + \gamma\,\mathrm{SPEI}_{it} + \alpha_i + \tau_t + \varepsilon_{it},$$

with sub-watershed (α_i) and year (τ_t) fixed effects and entropy-balancing weights. Because treatment is time-invariant, the dose is meteorological SPEI rather than calendar year: the year fixed effects absorb the common megadrought shock, and β is the differential deficit-to-impact slope between dammed and matched-control basins, robust to the siting-in-arid-central-Chile confound, which biases levels far more than slopes. The vulnerability hypothesis predicts $\beta > 0$.

Equivalence testing and minimum detectable effect. Because the headline results are nulls, we move beyond failure-to-reject and bound the operational effect. For each null we report a two one-sided test (TOST): the effect is declared equivalent to zero if its 90% confidence interval lies within a negligible region of ±25% of the baseline (untreated) transmission slope. We are not aware of an established hydrological or operational standard defining a "negligible" reservoir effect on drought transmission, so we anchor the margin in the granularity of operational drought classification rather than choosing it arbitrarily. Standardized drought indices are managed in discrete severity categories each 0.5 standardized units wide[28,36]. Because the estimand is a slope, the absolute change it induces in the propagated index scales with the SPEI deficit, so the relevant test is at the most extreme drought actually observed. The largest SPEI-12 deficit in the panel is ≈ -2.0 (minimum -1.98), and 25% of the baseline transmission slope is $0.25 \times 0.585 \approx 0.146$ SSI per unit SPEI (0.585 is the untreated baseline of the intent-to-treat panel; the downstream-gauge panel baseline is 0.531, and both appear in Supplementary Table S2); the maximum absolute perturbation of the streamflow-drought index a 25% slope alteration can produce is therefore $0.146 \times 1.98 \approx 0.29$ SSI units, below the 0.5-unit width of a single severity category. A 25% slope change thus cannot move a basin across an operational drought-severity threshold even under the worst observed deficit, and is marginal for drought management. We stress that 25% of the transmission slope nonetheless corresponds to a non-trivial absolute water volume: translating the maximal margin perturbation (0.29 SSI units) through each treated downstream gauge's flow-to-SSI sensitivity gives a median of roughly 58 hm^3 over a 12-month season (IQR 16 to 137 hm^3 across the 40 gauges, about 11% of mean annual flow; the linear sensitivity overstates the dry-tail volume, so these are upper-end figures; Supplementary Table S16). A volume of this order is genuinely material for local water management, so the margin is anchored to the operational drought-severity classification, not to volumetric negligibility, and the causal weight of the null rests on the within-basin placebo rather than on the equivalence verdict. We adopt $\pm 25\%$ as this deliberately strict, pre-specified margin and, because the equivalence verdict is nonetheless sensitive to the bound, also report the threshold-free MDE so that readers may apply any margin they consider material. The qualitative conclusion (only large effects are excluded; no outcome is equivalent at $\pm 25\%$) does not depend on the exact margin. The minimum detectable effect (at 80% power, two-sided $\alpha = 0.05$) and the equivalence interval are built from the **standard deviation of the permutation-null distribution**, not the analytic standard error, since cluster-robust SEs over-reject at ~21 clusters (archived analysis code). To confirm the null is informative rather than underpowered, we additionally run a **positive control**: a known buffering slope is injected into the treated units and the full estimator and randomization inference are re-run, verifying that the design recovers an effect of policy-relevant size and correctly flags none when none is injected.

Siting-confound decomposition. To locate where any apparent buffering comes from, we fit a ladder on the same panel: (1) a naive pooled slope gap (no weights, no fixed effects, what a conventional dammed-vs-undammed study reports); (2) unit + time fixed effects, unweighted; (3) the design estimator (ebal weights +

fixed effects); (4) the design estimator with the baseline SPEI-to-outcome slope allowed to vary by Köppen region. It is important that these are distinct controls. The entropy balancing exact-matches *basin size and elevation* within Köppen main group; it does **not** balance the residual aridity gradient (which varies within B and C) nor impose a region-specific dose-response. The apparent buffering can therefore survive rungs (1) to (3), whose weighting leaves the within-region aridity gradient intact, and die only at rung (4), which lets each climate region have its own baseline transmission and so removes the between-region contrast that the matching does not. The collapse from (3) to (4) is the between-region aridity confound; the residual at (4) is the regulation estimate.

Within-basin upstream/downstream placebo. The primary selection-versus-effect discriminator contrasts the SPEI-12→SSI-12 transmission slope at regulated **downstream** reaches against physically unregulated **upstream** reaches of the *same* dammed basins, which share climate, geology, and selection history. A genuine regulation effect implies a downstream-specific dose-response that the upstream placebo does not reproduce; an attenuation equally present upstream is siting, not regulation. Of the 21 treated matched basins, 15 have both an upstream and a downstream gauge, giving near-national coverage rather than a single case study. The contrast is also split by Köppen region to test whether a regulation effect emerges in the high-supply south. Because upstream gauges sit at higher elevations than downstream gauges, we test the placebo's comparability assumption directly: we quantify the elevation gap and regress the per-gauge transmission slope on elevation among undammed control gauges, so that any elevation-driven regime difference is measured rather than assumed.

Inference

With only ~21 treated clusters, cluster-robust standard errors over-reject, so the primary inference is **randomization inference**: the treatment label is permuted among units within Köppen $\times$ aridity- tercile strata (999 permutations) and the observed statistic is compared against the resulting null distribution; cluster-robust confidence intervals are reported alongside for completeness. Because hydrological and land-use outcomes can be spatially autocorrelated, we assess residual spatial dependence for each outcome with Moran's I (great-circle distances, five-nearest-neighbour row-standardized weights, tested by 999 residual permutations). Where it is present, notably the water-rights outcomes ($I \approx 0.4$ to 0.6, $p = 0.001$; Supplementary Table S9), the unit-level stratified permutation can be anti-conservative because it does not preserve spatial structure. We therefore re-run the affected outcomes with a spatially-restricted permutation that swaps treatment at the watershed-block level, so co-located sub-watershed move together, and with watershed-block-clustered confidence intervals; this spatially-valid inference governs where the naive and spatial tests disagree (Supplementary Table S9). Parallel pre-trends are checked with a dynamic event study (dammed × year, reference 2009), and residual confounding is probed with an aridity placebo that relabels the most-arid control tercile as pseudo-treated.

Software and reproducibility

All analyses are implemented in R within a `targets` pipeline (matched-set construction, estimators, and figures are individual targets), using `WeightIt`/`cobalt` for balancing and diagnostics, `fixest` for fixed-effects regression, `sf`/`terra`/`exactextractr` for the spatial extractions, and `renv` for package pinning. Every result in the manuscript corresponds to a named pipeline target; no figure or table is produced by hand.

Data and code availability

The full analysis pipeline, comprising the `targets` workflow, the reusable R functions (`src/R/`), the study-parameter configuration (`config/`), and the scripts that generate every figure and table, is openly available at https://github.com/frzambra/dam_drought_effectiveness and archived on Zenodo (DOI: 10.5281/zenodo.XXXXXXX). The pinned R environment (`renv.lock`) reproduces the package versions used. Reservoir storage records, the derived result tables, and matched-set metadata are included in the repository; the third-party input datasets are publicly available from their providers as cited: CHIRPS/CHIRTS (SPEI-12 forcing and baseline aridity), MODIS MOD16 (evapotranspiration), MapBiomas Chile (land cover), the CR2 network (streamflow), ERA5-Land (snow water equivalent), SRTM (elevation), and the DGA reservoir, watershed, water-rights, and well-hydrograph registries. Every result in the manuscript corresponds to a named pipeline target, so the analyses can be regenerated with a single `targets::tar_make()`.

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